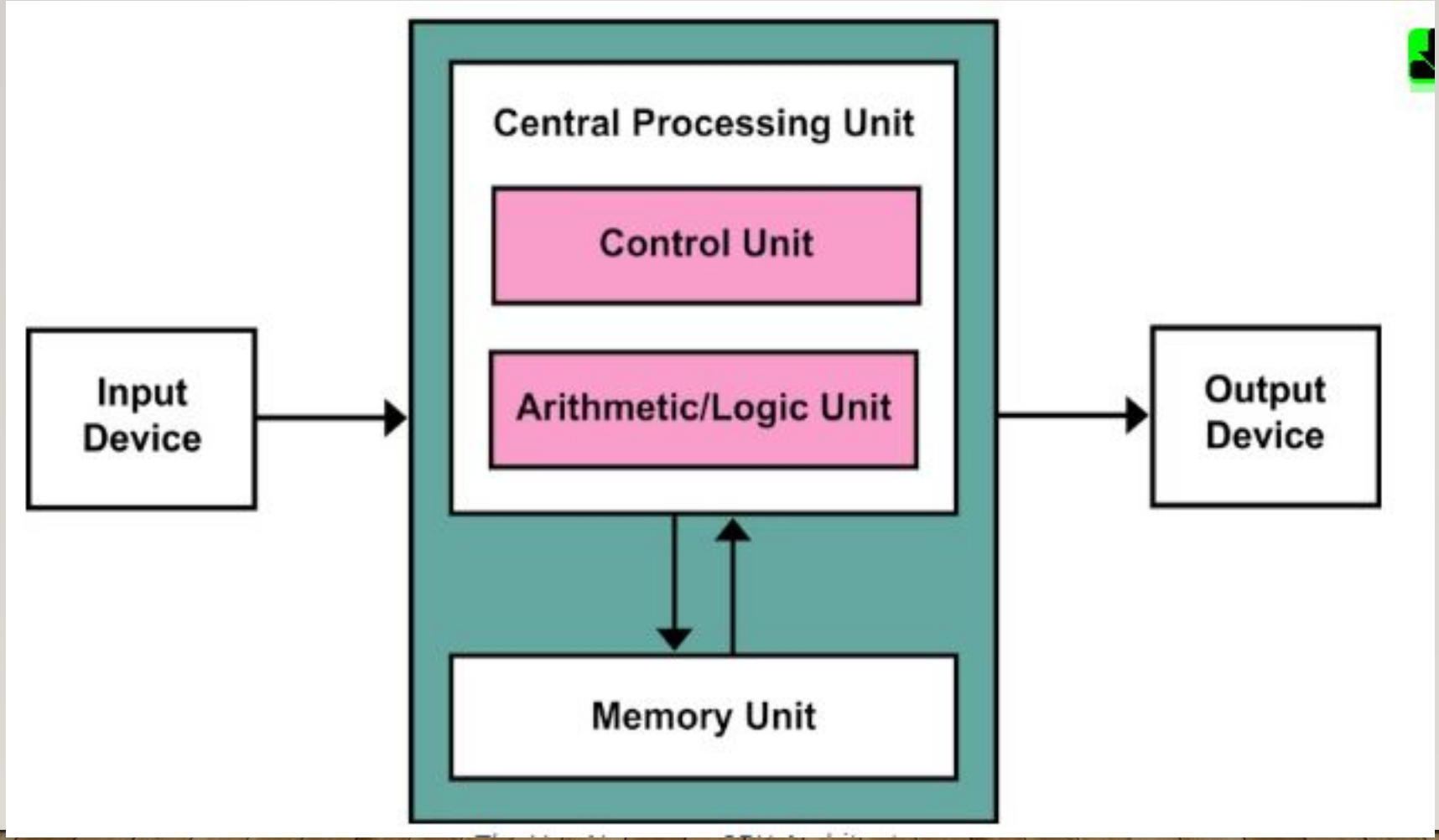
Modern CPUs

Understanding Modern CPU Architecture:



- The Central Process Unit is usually referred to as "The Brain" of the computer. The CPU is at the heart of all data interpretation and processing operations in a computer.

- The CPU contains three major components: memory, control unit, and the ALU(arithmetic and logic unit).

- The modern CPU architecture is implemented on ICs(Integrated Circuits), with one or two single metal-oxide-semiconductor IC chips.

- Memory: A memory component for storing data and instructions
- Central Processing Unit: for decoding and executing instructions
- Inputs/Outputs: and a set of inputs and output interfaces

A microprocessor is an integrated circuit containing a CPU, responsible for executing instructions in a computer. The CPU is the core processing unit, handling computations and control functions.

- A chip's architecture determines where its parts are located and connected, how it connects with other parts of the computer, and much more.
- A processor's architecture determines how many transistors it has, and therefore the processor's power

- A processor includes many other features that affect its performance. For example, a processor's performance is affected by the number of bits of data it can process at any one time.
- In 2003 American Micro Devices (AMD) released a new generation of desktop PC processor that can handle 64 bits of data.

- A 64-bit processor refers to a microprocessor that can process data and instructions in chunks of 64 bits.

Look Inside the CPU:

- Architecture:

  Determines
- Location Of CPU.
- Bit size
- determines how many transistors it has

Intel:

- Intel is historically the leading provider of chips for PCs.
- 1971, Intel invented the microprocessor the so-called computer on a chip—with the 4004 model.
- IBM pc powered by intel 8086.
- Current processors:

a) Xeon.
b) Core i7, i5, i3
c) Pentium.

Microcomputer Processors:

- Advanced Micro Devices(AMD)
- Main competitor to intel.
- Originally produced budget products
- Current products outperform Intel
- Current processors

Opteron.
FX.
Athlon.

Freescale:

- A Subsidiary of Motorola.
- Co-Developed the Apple G4 PowerPC
- Currently focus on linux Market

800 MHz Motorola PowerPC 7450 processor on the CPU module of a Power Mac G4

IBM:

- Historically manufactured mainframes.
- Partnered with Apple to develop G5
  First Consumer 64 bit

Comparing Processors:

- Speed of processor.
- Size of cache.
- Number of registers.
- Bit size.
- Speed of side bus.

Advanced Processor Topics:

- **RISC Processors**

 Reduced Instruction Set Computing.
 Smaller instruction sets.
 May process data faster.
 PowerPC and G5.

- **CISC Processors**

  Complex Instruction Sets Computing.
 Large instruction sets.
 More Applications.
 IBM Compatible.

Parallel Processing

- Multiple processors in a system.
- Symmetric Multiple Processing
  - Number of processors is a power of 2.

- Massively Parallel Processing.
  - Thousands of processors.

Standard computer port:

- Keyboard and mouse ports.
- USB ports.
- Parallel ports.
- Network ports.
- Modem ports.
- Audio ports.
- Serial ports.
- Video ports.

Extending The Processors Power

- **SCSI**

- Small Computer System Interface.
- Supports of dozens of devices.
- External devices daisy chain.
- Fast hard drives and CD-ROMs.

USB

- Universal Serial Bus.
- Most popular external.
- Supports up to 127 devices.
- Hot swappable.

FireWire:

- IEEE 1884
- Cameras and video equipment.
- Hot swappable.
- Port is very expensive.

Expansion slots and boards:

- Allows users to configure the machine.
- Slots allow the addition new devices
- Devices are stored on cards.
- Computer must be off before inserting.

Plug and play:

- New hardware detected automatically .
- Prompts to install drives.
- Non-technical users can install devices.